Course Objectives

The aim of this course is to provide students with:

- The basic knowledge of different type of oscillations and waves and there real- life applications.

- The fundamentals of physical optics including interference and diffraction.

Force, Work and Energy

Linear Motion and the Concept of Force

- Motion is defined as the change in the position of the object with time due to an applied force.
- Motion is classified into *linear motion* where the object moves in a straight line, and *circular motion* where the path is circular.

The main parameters defining the linear motion are:

- ***Displacement***: It is the change in particle position (at time t) from an initial position P_0.
- ***Velocity***: It is the time rate of change of the linear displacement,

$$\vec{v} = \frac{d\vec{x}}{dt}$$

- ***Acceleration:*** is defined as the time rate of change of the linear velocity.

$$\vec{a} = \frac{d\vec{v}}{dt}$$

Acceleration is a vector quantity. Notice that the velocity vector $\vec{v}$ changes if its magnitude or its direction or both change and in any of such cases a linear acceleration is produced. The direction of the velocity of an object and its acceleration are related as follows; they are both in the same direction if the object is speeding up and in opposite direction if the object is slowing down (deceleration)

Force

Momentum: it is equal to the product of the mass and the velocity of the object.

$$\vec{p} = m\vec{v}$$

The net force acting on a particle equals the time rate of change of the momentum of the particle**.**

$$\sum \vec{F} = \frac{d\vec{p}}{dt} = \frac{d(m\vec{v})}{dt}$$

In case that m is constant during the motion last equation reduced to

$$\sum \vec{F} = \frac{md\vec{v}}{dt} = m\vec{a}$$

Note: the mass of a rocket changes as fuel is burned and ejected from the rocket. In such case we must use a momentum approach

Forces can be divided into two main categories:

Contact forces

This type of forces involve physical contact between two objects.

Examples:

- push or pull an object
- A player kicking a ball
- the force exerted by gas molecules n the walls of a container.

Field forces

These are long-range forces, they do not involve physical contact between two objects. They act through empty space.

Examples:

- the force of gravity between the masses of two objects
- the attraction force between a magnet and an iron bar
- the electric force that one electric charge exerts on another

Examples:

1. The momenta of two moving particles are equal. This means that:

a) The two particles have equal masses m

b) The two particles have equal speeds v

c) The two particles can only be stopped when acted upon by the same force.

d) The two particles have the same kinetic energy

Ans: c

2. The aside figure shows the velocity-vs-time graph for a basketball player traveling up and down the court in a straight-line path.

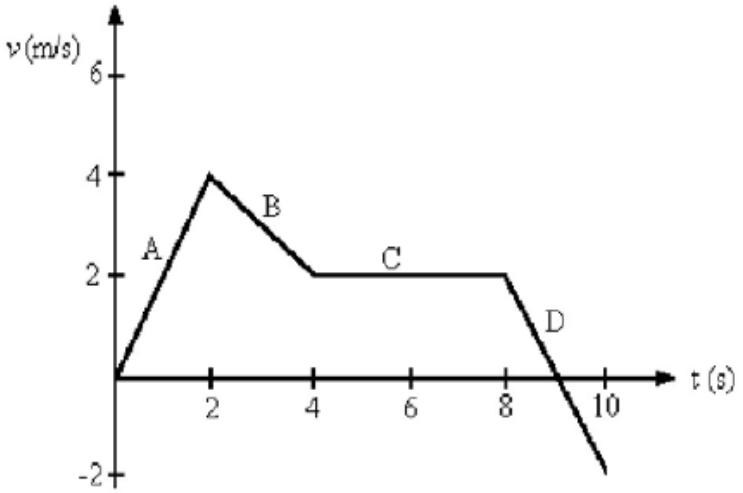
Find the net displacement of the player for the 10 s shown on the graph

a) 20 b) 18

c) 16 d) 14

e) 12

Ans: b



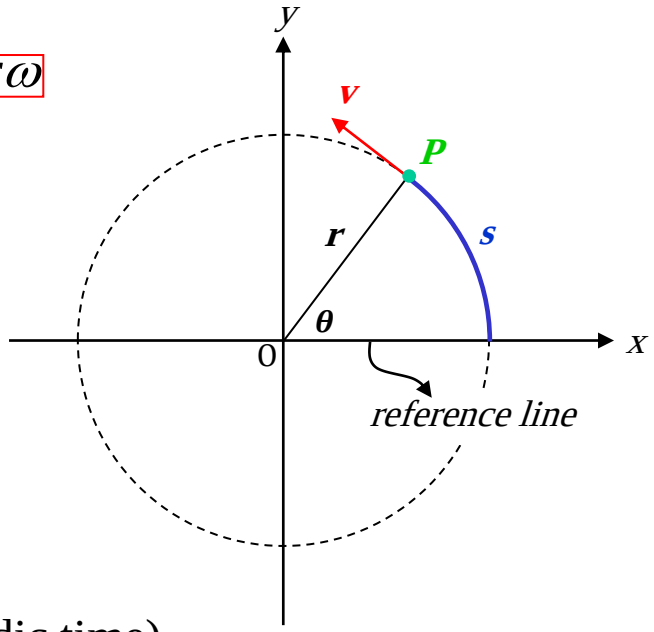
Uniform Circular Motion

During the time interval *t,* a particle (*P*) moves from the *reference line (x-axis)* and covers a linear distance (*s*), while the radius vector covers an angular distance (θ).

by definition: $v\ (linar\ speed) = \frac{ds}{dt}$ and $\omega(angular velocity) = \frac{d\theta}{dt}$

$$\theta = \frac{s}{r} \quad or \quad s = r\theta \quad \Rightarrow \quad \frac{ds}{dt} = r\frac{d\theta}{dt} \quad \Rightarrow \quad \boxed{v = r\omega}$$

A particle moving in a circle of radius *r* with constant speed *v* is in a ***uniform circular motion***



$$\boxed{\omega = \frac{d\theta}{dt} = \frac{v}{r} = \text{constant}}$$

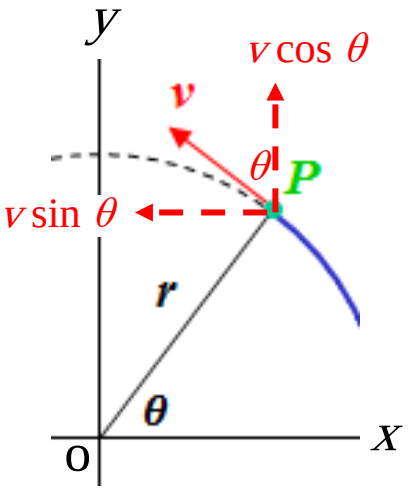
Thus, $\int_0^{\theta} d\theta = \int_0^{t} \omega dt = \omega \int_0^{t} dt \quad \Rightarrow \quad \theta = \omega t$

For *one complete revolution*: $\theta \rightarrow 2\pi$ & $t \rightarrow T$ (periodic time)

$$T = \frac{2\pi}{\omega} = \frac{2\pi r}{v} \qquad \text{and} \qquad f\ (frequency) = \frac{1}{T} = \frac{\omega}{2\pi} = \frac{v}{2\pi r}$$

For uniform circular motion, ***the linear velocity vector*** $(\vec{v})$ has constant magnitude and is changing in direction.

From the next figure $\vec{v} = -(v \sin\theta)\vec{i} + (v \cos\theta)\vec{j}$



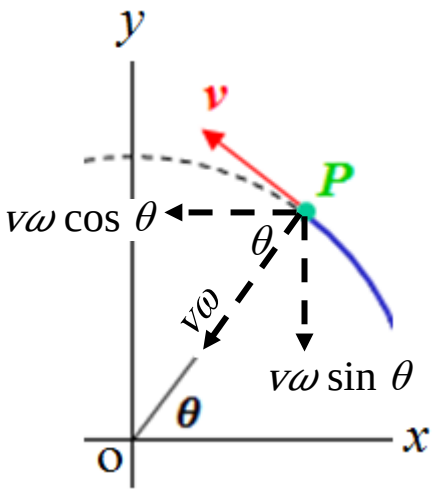
then acceleration is given by:

$$\vec{a} = \frac{d\vec{v}}{dt}$$

$$= -\left(v\cos\theta\frac{d\theta}{dt}\right)\vec{i} - \left(v\sin\theta\frac{d\theta}{dt}\right)\vec{j}$$

$$= -(v\omega\cos\theta)\vec{i} - (v\omega\sin\theta)\vec{j}$$

and is directed from P towards O, the center of rotation, see next figure. For this reason this type of acceleration is called the ***centripetal (radial) acceleration***.



The magnitude (a_c) of this acceleration is given by:

$$a_c = \sqrt{v^2\omega^2\cos^2\theta + v^2\omega^2\sin^2\theta} = v\,\omega = \frac{v^2}{r}$$

- If v is variable, then there is another component for acceleration (tangential acceleration),

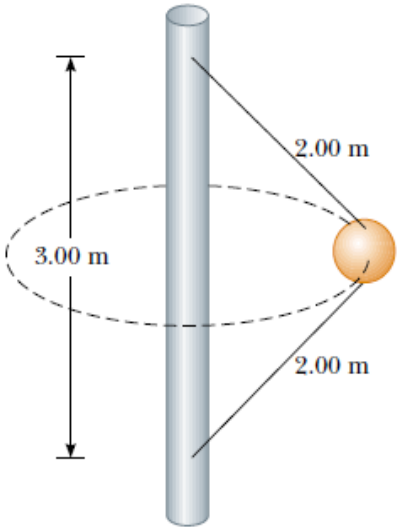
$$a_t = \frac{dv}{dt} = r\frac{d\omega}{dt} = r\alpha$$

Kinematic Equations for Rotational and Translational Motion

Constant linear acceleration $\boldsymbol{a}$	Constant angular acceleration α
$v = v_0 + at$	$\omega = \omega_0 + \alpha t$
$x = v_0 t + \frac{1}{2}at^2$	$\theta = \omega_0 t + \frac{1}{2}\alpha t^2$
$v^2 = v_0^2 + 2ax$	$\omega^2 = \omega_0^2 + 2\alpha\theta$

- Example 3: A $4\ kg$ object is attached to a vertical rod by two strings, as shown in Figure. The object rotates in a horizontal circle at constant speed $6\ m/s$. Find the tension in a) the upper string and b) the lower string

Solution:



$$W_g = mg = 4 \times 9.8 = 39.2\ N$$

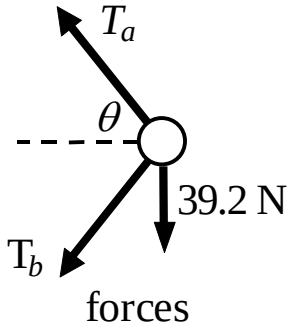
$$\sin\theta = \frac{1.5}{2} \quad \rightarrow \quad \theta = 48.6^o$$

$$r = 2\cos 48.6^o = 1.32\ m$$

$$\sum F_x = ma_x = ma_c = \frac{mv^2}{r}$$

$$T_a \cos 48.6^o + T_b \cos 48.6^o = \frac{4 \times (6)^2}{1.32}$$

$$T_a + T_b = 165\ N \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (1)$$



forces

$$\sum F_y = ma_y = 0$$

$$T_a \sin 48.6^o - T_b \sin 48.6^o - mg = 0$$

$$T_a - T_b = 52.3\ N \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (2)$$

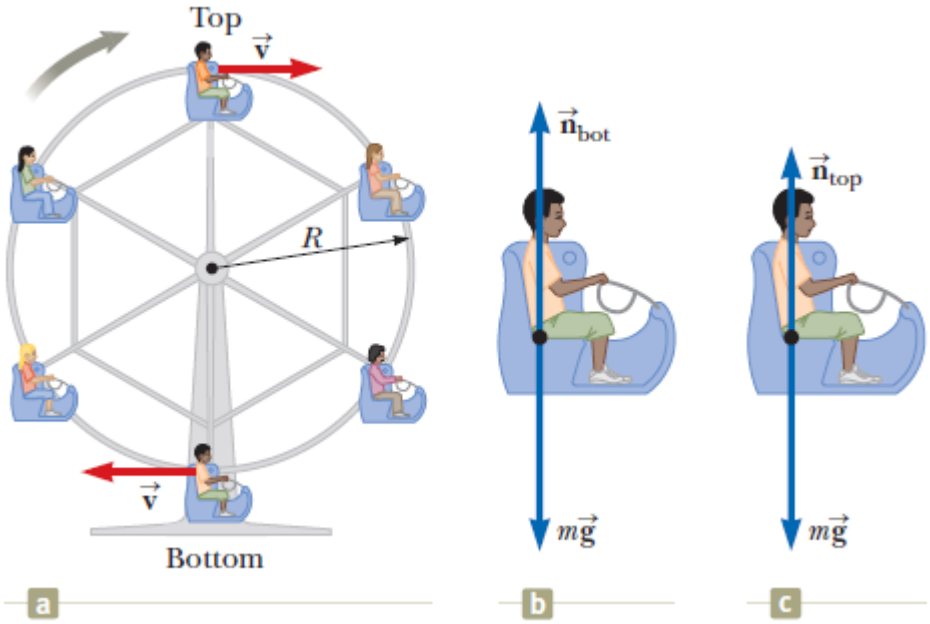
By solving equations (1) and (2) simultaneously, we find that,

$$T_a = 108\ N \qquad T_b = 56.2\ N$$

Example Riding the Ferris Wheel

A child of mass m rides on a Ferris wheel as shown in Figure 6.6a. The child moves in a vertical circle of radius 10.0 m at a constant speed of 3.00 m/s.

(A) Determine the force exerted by the seat on the child at the bottom of the ride. Express your answer in terms of the weight of the child, mg.



Analyze We draw a diagram of forces acting on the child at the bottom of the ride as shown in Figure 6.6b. The only forces acting on him are the downward gravitational force $\vec{\mathbf{F}}_g = m\vec{\mathbf{g}}$ and the upward force $\vec{\mathbf{n}}_{\text{bot}}$ exerted by the seat. The net upward force on the child that provides his centripetal acceleration has a magnitude $n_{\text{bot}} - mg$.

$$\sum F = n_{\text{bot}} - mg = m\frac{v^2}{r}$$

$$n_{\text{bot}} = mg + m\frac{v^2}{r} = mg\left(1 + \frac{v^2}{rg}\right)$$

$$n_{\text{bot}} = mg\left[1 + \frac{(3.00\ \text{m/s})^2}{(10.0\ \text{m})(9.80\ \text{m/s}^2)}\right]$$

$$= 1.09\,mg$$

Hence, the magnitude of the force $\vec{\mathbf{n}}_{\text{bot}}$ exerted by the seat on the child is *greater* than the weight of the child by a factor of 1.09. So, the child experiences an apparent weight that is greater than his true weight by a factor of 1.09.

(B) Determine the force exerted by the seat on the child at the top of the ride.

SOLUTION

Analyze The diagram of forces acting on the child at the top of the ride is shown in Figure 6.6c. The net downward force that provides the centripetal acceleration has a magnitude $mg - n_{top}$.

Apply Newton's second law to the child at this position:
$$\sum F = mg - n_{top} = m\frac{v^2}{r}$$

Solve for the force exerted by the seat on the child:
$$n_{top} = mg - m\frac{v^2}{r} = mg\left(1 - \frac{v^2}{rg}\right)$$

Substitute numerical values:
$$n_{top} = mg\left[1 - \frac{(3.00\ \text{m/s})^2}{(10.0\ \text{m})(9.80\ \text{m/s}^2)}\right]$$
$$= 0.908\,mg$$

In this case, the magnitude of the force exerted by the seat on the child is *less* than his true weight by a factor of 0.908, and the child feels lighter.